



The value of 24-hour video-EEG in evaluating recurrence risk following a first unprovoked seizure: A prospective study



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ABSTRACT

Purpose: To evaluate the value of 24-hour video-EEG (VEEG) in assessing recurrence risk after a first unprovoked seizure.

Method: Consecutively 134 patients with a first unprovoked epileptic seizure were recruited from West China Hospital, Sichuan University, between January 2010 and January 2013. All patients underwent VEEG and magnetic resonance imaging (MRI) of the brain. Each patient had at least 24-month follow up.

Results: Seventy-six (56.7%) patients had abnormal VEEG, and VEEG abnormalities were associated with an increased risk of seizure recurrence (RR 2.84, 95% CI 1.67–4.82, $p < 0.001$). The overall accumulated seizure recurrence risks were 51.5% in all patients, and 45.6% in generalized seizures subgroup, and with no significant difference. The subgroup of VEEG with epileptiform discharges had an increased seizure recurrence risk compared with normal VEEG (RR 2.76, 95% CI 1.83–5.34, $P < 0.001$) and the nonsignificant abnormality VEEG group (RR 2.05, 95% CI 1.14–3.82, $P < 0.001$). Within the group of whom showed epileptiform discharges, the recurrence rate of those with generalized epileptiform discharge abnormality and focal epileptiform discharge abnormality were not significantly different (RR 1.09, 95% CI 0.44–2.69, $P = 0.85$).

Conclusions: An abnormal VEEG is a risk factor for seizure recurrence in patients with a first unprovoked seizure, especially if epileptiform discharges past. The recurrence risks were 73.2% in the epileptiform discharges abnormality VEEG group, which may help the diagnosis of epilepsy according to the practical clinical definition of epilepsy.

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1. Introduction

A seizure is a clinical manifestation characterized by abnormal excessive or synchronous neuronal activity in the brain that occurs instantaneously [1]. Seizures are categorized into provoked seizure and unprovoked seizure, according to whether there are one or more predisposing factors [2]. A provoked seizure is defined as an acute occurrence caused by stroke, traumatic brain injury, intracranial surgery, central nervous system (CNS) infections or other acute brain disease. And an unprovoked seizure is of unknown cause. The risk of a single seizure occurring during the lifespan in the general population is 5–10% [2,3], and approximately 3% of these cases may develop into epilepsy [4].

The definition of epilepsy includes a conceptual definition and a practical definition in epilepsy research. A conceptual definition of

epilepsy was formulated by the International League Against Epilepsy (ILAE) and the International Bureau for Epilepsy (IBE) in 2005. According to this conceptual definition, epilepsy is a disorder of the brain that involves at least one epileptic seizure [1] and usually involves two unprovoked seizures that occur at least 24 h apart [5]. In 2014, the ILAE formulated a practical clinical definition of epilepsy [6], with one of the diagnostic criteria specifying the occurrence of one unprovoked (or reflex) seizure with a recurrence risk of at least 60% over the next 10 years. According to this new practical definition, the following situations may be diagnosed as epilepsy: a patient with a single seizure occurring at least a month after a stroke [6,7] or a child with a single seizure, an epileptiform EEG and a structural or remote symptomatic etiology [6,8].

However, the new definition stating that a single seizure with epileptiform EEG could be diagnosed as epilepsy conflicts with the results of previous studies that have shown contradictory results for the predictive value of the EEG for the risk of seizure recurrence. Most of the previous studies have evaluated only the value of standard EEG for predicting the recurrence risk after the first

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unprovoked seizure. Few studies have investigated the EEG over a long time period [9–15]. The purpose of our prospective study was to evaluate the value of 24-hour VEEG in assessing the recurrence risk after the first unprovoked seizure.

2. Materials and methods

Clinic patients or outpatients who had experienced a single first unprovoked seizure were recruited from the West China Hospital of Sichuan University between January 2010 and January 2013. Two well-trained and experienced epileptologists identified patients who had experienced a first paroxysmal event and decided whether to include these patients based on the description of the seizure by witnesses. The diagnostic criteria for seizure were according to those formulated by the ILAE in 2005 [1]. All seizure types were categorized as generalized and partial onset according to the classification of seizures published by the ILAE in 1981 [16].

In our study, first unprovoked seizures included partial seizures (simple, complex and secondarily generalized tonic-clonic seizures) and generalized seizures (generalized tonic-clonic seizures, tonic seizures and clonic seizures). Patients were also included in our study if their first seizures were status epilepticus and two or more seizures had occurred within 24 h (cluster seizures). Patients were excluded from our study if their first paroxysmal events were non-epileptic seizures, acute symptomatic seizures (associated with acute factors, such as acute period of stroke, brain injury or central nervous system infections, abrupt withdrawal of alcohol, acute intoxication), other seizure types (absence, myoclonic seizures and unclassified epileptic seizures), and others who reluctant to participate our study (Fig. 1).

Upon study enrollment, patients received examinations including full general and neurological examinations, standard blood and serum electrolyte laboratory tests, three-dimensional MRI, and a

24-hour VEEG. All relevant clinical information and demographic data were acquired and included the following: sex, age, location, history of alcohol and tobacco use, marital status, educational level and profession, seizure type, presence/absence of nocturnal seizure, and any detrimental items in the medical history that may cause seizure or increase recurrence risk before their first unprovoked seizures (e.g., dystocia, febrile seizure, cerebral trauma, brain surgery, stroke, CNS infections, and family history of epilepsy).

2.1. VEEG

A 24-hour VEEG (EBNeuro S.p.A, Germany) was recorded for each patient within 7 days after the first unprovoked seizure. We used 24-channel digital recordings (international standard 10/20 system) with bipolar and average reference montages (time constant adjusted to 0.3, tracing speed to 30 mm/s, frequency filter to 70 Hz, and sensitivity between 7 and 10 V/mm). Two EEG experts who were blinded to the study analyzed the 24-hour VEEG results.

The VEEG recordings were classified as either normal or abnormal. An abnormal VEEG was further categorized as either (a) non-epileptiform discharges (e.g., increased slow waves, bilateral asymmetry or disappearance of the normal sleep wave) or (b) epileptiform discharges (e.g., spikes or sharp waves, or spike and wave discharges) that were generalized or focal discharges. If VEEG were with both generalized and focal epileptiform discharges or focal secondarily generalized epileptiform discharges, we considered them to be focal.

2.2. MRI

A three-dimensional MRI was performed on 1.5 or 3.0 tesla MAGNETOM Avanto scanners (Siemens AG, Germany). MRI scans

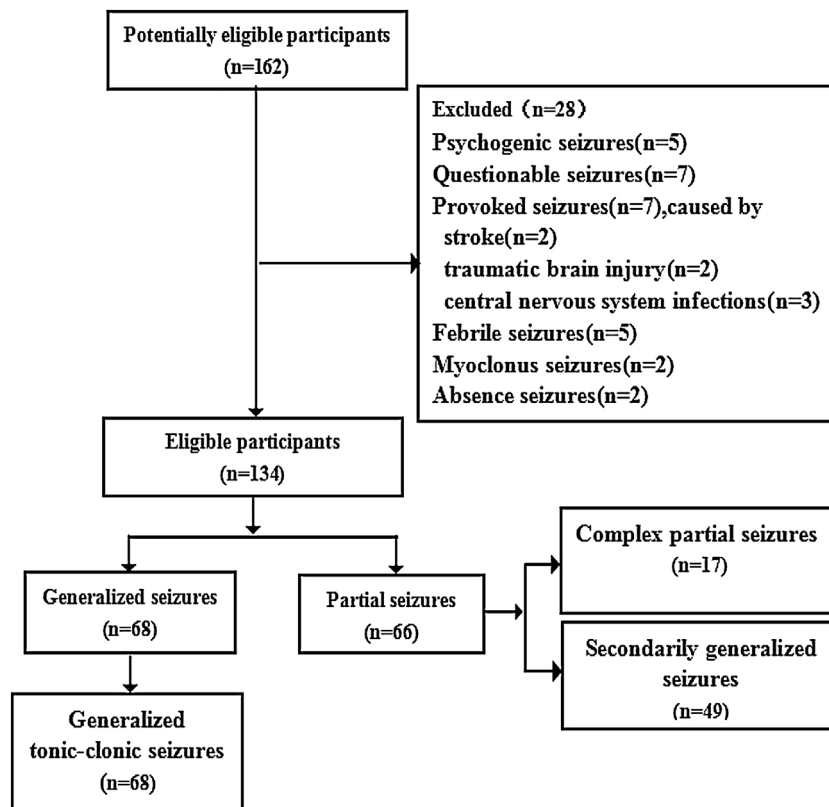


Fig. 1. Flowchart showing inclusion and exclusion of our study.

were reviewed by a radiologist and an epileptologist and divided into normal and abnormal according to whether any observed foci could be responsible for seizure onset or not. Abnormal MRI scans included leukomalacia, gliosis, encephalomalacia, any gray matter lesion, mass lesion, hemorrhage, vascular lesion, hippocampal abnormality, ventricular enlargement greater than 1.5 cm, or prominence of extra-axial fluid spaces greater than 1.0 cm [10].

2.3. Follow-up

The Institutional Ethics Committee of West China Hospital gave ethical approval for our study. Each patient had given informed consent prior to the study and agreed not to receive anti-epileptic drugs (AEDs) treatment until seizure recurrence. The patients came to our outpatient clinic at follow-up times of 3, 6, 12, and 24 months after a first unprovoked seizure. We attempted to maintain contact with patients by phone or letter if they failed to return to our clinic. If a seizure recurred, the follow-up was ceased, and then the patient began receiving AEDs treatment. For the patients without recurrent seizures, the follow-up ultimately lasted until January 2015, ensuring that the shortest follow-up time was at least 2 years.

2.4. Statistical analysis

SPSS 21.0 software (IBM, Chicago, IL, USA) was used to analyze the data from all patients who had a first unprovoked seizure. We used the descriptive analyses of mean, median, mode, and standard deviation. We used the chi-square test (T test) and logistics to evaluate whether relative risk factors were predictive factors for seizure recurrence in the bivariate analyses. Survival analysis was performed using Kaplan-Meier survival curves. The COX proportional hazard model was used in univariate analyses. Statistical significance was established at the 5% level ($p < 0.05$). Fig. 2 shows the analysis route of our study.

3. Results

Our study included 134 patients, of whom 89 (66.4%) were males and 55 (33.6%) were females. The patient ages ranged from 3 to 77 years when the first unprovoked seizure occurred. The mean age ($\pm$ SD) at first seizure was 25.8 ± 6.0 years. There were 29 patients (21.6%) with a detrimental medical history that may have caused the seizure or increased the recurrence risk before their first unprovoked seizures as follows: 7 patients (5.2%) with traumatic brain injuries, 6 (4.5%) with cerebrovascular accidents, 5 (3.7%) with encephalitis, 8 (6.0%) with febrile seizure, 2 (1.5%) with family history of epilepsy in immediate family members (both had mothers with epilepsy), and 1 (0.8%) with dystocia. Fifteen patients (11.2%) had their first seizure during sleep (nocturnal seizure), and 9 patients (6.7%) had multiple seizures or presented as status epilepticus within 24 h. Finally, the percentages of patients with generalized seizures or partial seizures (including secondary generalized) were 50.7% and 49.3%.

None of the patients experienced seizure recurrence during 24-hour VEEG monitoring. The interictal VEEG was normal in 58 patients (43.3%) and abnormal in 76 (47.7%) patients. The 76 patients with abnormal VEEGs included 20 (26.3%) patients with a nonsignificant abnormality and 56 (73.7%) patients with an epileptiform discharge abnormality. The epileptiform discharge abnormalities were generalized epileptiform discharges in 16 (21.1%) patients and focal epileptiform discharges in 40 (52.6%) patients.

Three-dimensional MRI scans of the head were normal in 101 patients (76.4%) and abnormal in 33 (24.6%) patients. The abnormal MRI scans showed 7 cases of encephalomalacia, 4 brain contusions, 4 infarctions, 2 cerebral hemorrhages, 6 cases of hippocampal sclerosis, 6 cases of focal cortical dysplasia and 2 cases of other abnormalities.

Sixty-nine patients experienced seizure recurrence in our study. The overall accumulated risks of recurrence were 8.2%, 16.4%, 29.9%, 46.3%, and 51.5% at 3, 6, 12, 24 and over 24 months, respectively (Fig. 3). The overall product-limit estimate of

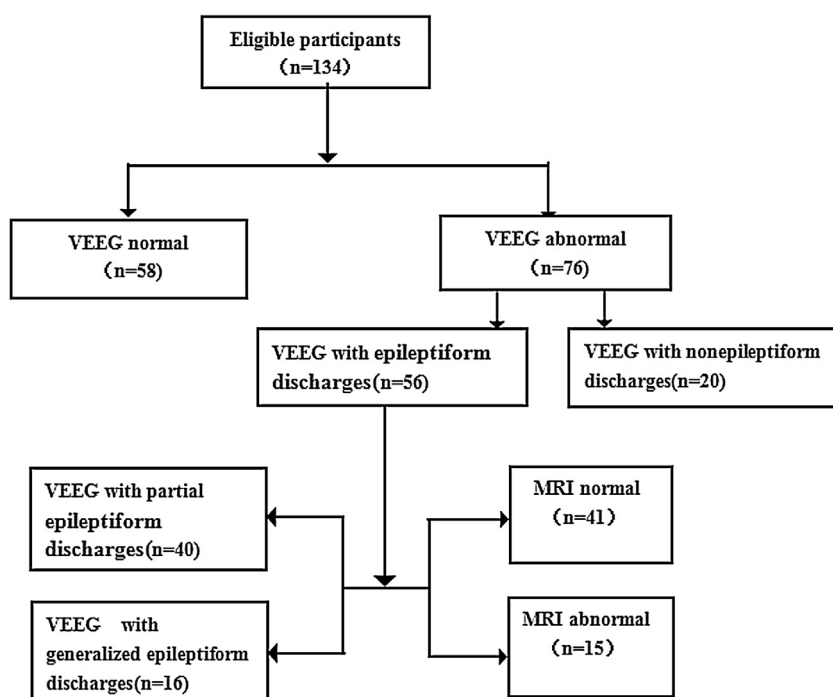


Fig. 2. Analysis route for Research roadmap of first unprovoked seizure.

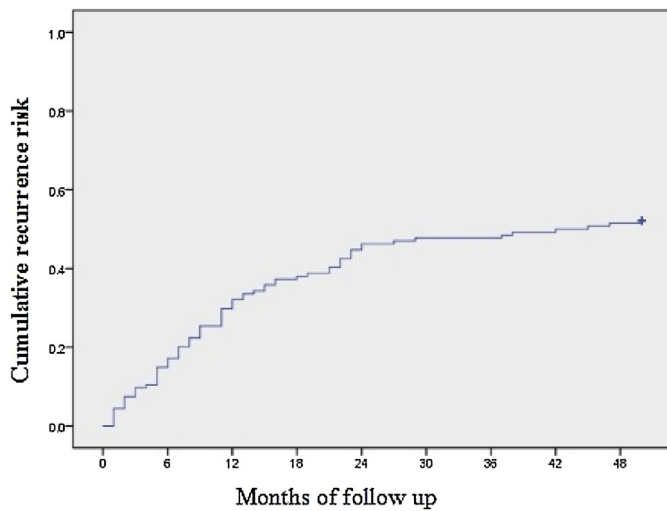


Fig. 3. Risk of recurrence risk after first unprovoked seizure: Kaplan-Meier curve.

recurrence was 61.9% at 6 months, 85.7% at 1 year, and 95.2% at 2 years after seizure onset.

Only a detrimental medical history (RR 1.80, 95% CI 0.98–3.30, $P = 0.03$) or an abnormal VEEG (RR 1.97, 95% CI 1.37–2.82, $P < 0.001$) were predictive factors for seizure recurrence in the bivariate analyses, as shown in Table 1. However, multivariate analysis using the Cox regression model determined that only an abnormal VEEG was a statistically significant predictor for seizure recurrence (RR 2.84, 95% CI 1.67–4.82, $P < 0.001$).

The overall accumulated recurrence risks were 45.6% (31/68) in generalized seizures subgroup, and the recurrence risks of patients with normal VEEG, nonsignificant abnormality VEEG and epileptiform discharge abnormality VEEG were 31.4%, 33.3%, 70.8%, respectively. Furthermore, the epileptiform discharge abnormality VEEG was associated with an increased seizure recurrence risk compared with the normal VEEG (RR 2.25, 95% CI 1.30–3.92, $P = 0.004$). The risk of recurrence was not different in generalized seizures subgroup compared to all patients who had a first unprovoked seizure ($P = 0.427$).

The recurrence risks were 32.8% in the normal VEEG group, 45.0% in the nonsignificant abnormality VEEG group and 73.2% in

the epileptiform discharge abnormality VEEG group. There was no statistically significant difference associated with seizure recurrence between the normal VEEG group and the nonsignificant abnormality VEEG group (RR 1.12, 95% CI 0.73–2.54, $P = 0.348$). However, the epileptiform discharge abnormality VEEG group was associated with an increased seizure recurrence risk, compared with the normal VEEG group (RR 2.76, 95% CI 1.83–5.34, $P < 0.001$) and the nonsignificant abnormality VEEG group (RR 2.05, 95% CI 1.14–3.82, $P < 0.001$), as shown in Fig. 4.

Subgroup analysis found that the seizure recurrence rates among generalized epileptiform discharges and focal epileptiform discharges were 68.8% and 75.0%, respectively, and were not significantly different (RR 1.09, 95% CI 0.44–2.69, $P = 0.85$), as shown in Fig. 5. However, an abnormal MRI scan was significantly associated with an increased risk of seizure recurrence (RR 1.46, 95% CI 0.87–4.72, $P = 0.046$) in patients with an abnormal VEEG that showed epileptiform discharge abnormalities (Fig. 6).

4. Discussion

The risk of recurrence following the first unprovoked seizure is a controversial topic in the field of epilepsy. Thus far, there are no definitive conclusions about the predictive value of the EEG after a first unprovoked seizure. Our study found that the rate of VEEG abnormalities was 56.7% in first unprovoked seizure patients, similar to previous studies that reported a rate of EEG abnormalities ranging from 45% to 84% in all first seizure patients [9,11–15], and only an abnormal VEEG was a statistically significant predictor for seizure recurrence. Previous studies have found that recurrence risk factors after a first seizure included a history of brain insult (such as strokes, cerebral hemorrhages, CNS infections, and traumatic brain injuries), EEG with epileptiform discharges, MRI with significant abnormalities, a partial seizure, or a nocturnal seizure [2–4,8–15,17,18,21,22]. The risk factor of seizure recurrence after the first unprovoked seizure confirmed in our study is 51.5%. This different of risk factors for recurrence in first unprovoked seizure patients between our study and others may be due to the characteristics of the study populations or methodological differences.

Table 1
Characteristics for demography and seizure, risks for recurrence.

Risk factors		Recurred/Total	P	RR	95%CI
Gender	Male	41/89(46.1%)	0.08	1.43	0.94–2.18
	Female	28/45(62.3%)			
Age	<18 years	29/54(53.7%)	0.67	1.12	0.58–2.31
	≥18 years	40/80(50.0%)			
Seizure type	Generalized	31/68(45.6%)	0.17	0.62	0.31–1.22
	Partial	38/66(57.6%)			
Detrimental history	Yes	19/27(70.4%)	0.03	1.8	0.98–3.30
	No	50/107(46.7%)			
Cluster seizure	Yes	2/9(22.2%)	0.07	4.04	0.81–20.23
	No	67/125(53.6%)			
Seizure period	Sleep	10/15(66.7%)	0.21	1.51	0.72–3.16
	Awake	59/119(49.6%)			
MRI	Normal	47/101(46.5%)	0.05	1.6	0.96–2.69
	Abnormal	22/33(66.7%)			
VEEG	Normal	19/58(32.8%)	<0.001	1.97	1.37–2.82
	Abnormal	50/76(65.8%)			

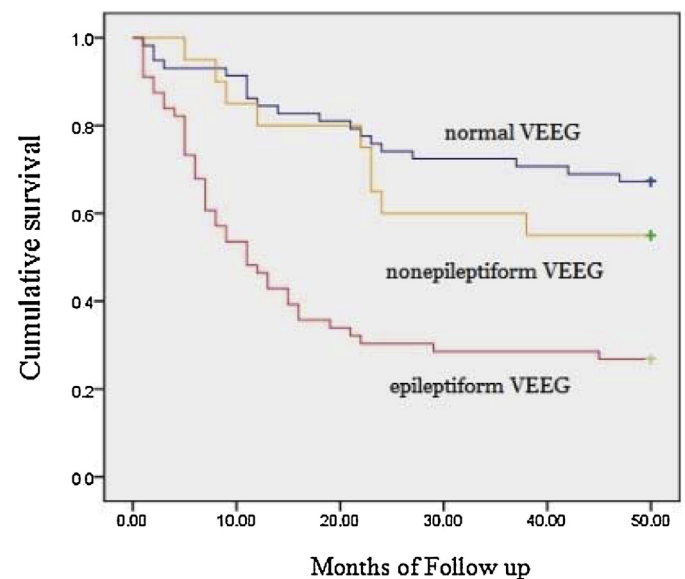


Fig. 4. Non-recurrence among patients with VEEG that were normal, had non-epileptiform discharges or had epileptiform discharges.

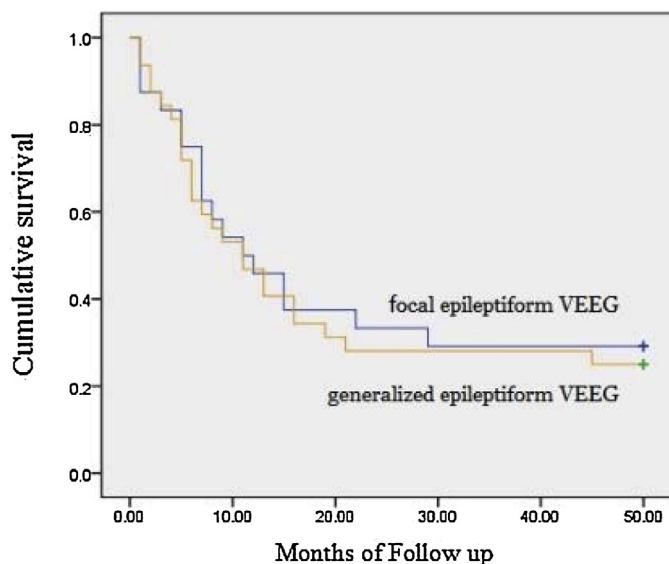


Fig. 5. Non-recurrence among patients with an abnormal VEEG that showed either focal epileptiform discharges or generalized epileptiform discharges.

The overall accumulated seizure recurrence risks were 51.5% in all patients, and 45.6% in generalized seizures subgroup, and with no significant difference. Recurrence risk rates after the first unprovoked seizure may vary in different studies, Prior studies estimate that they may range from 6% to 82% [18–20]. The recurrence risk in a prospective study with up to 2 years of follow-up after the first unprovoked seizure is 36% and a rate of 47% was found in a retrospective study [10]. Another recent evidence-based study concluded that the recurrence risk rate was 21–45% in the first 2 years in adults [19]. It is apparent that our result is similar to those in the studies mentioned above, even given our study limitations of a small number of patients (only 134) with a wide range in age (3–77 years) who came from different areas (both urban and rural).

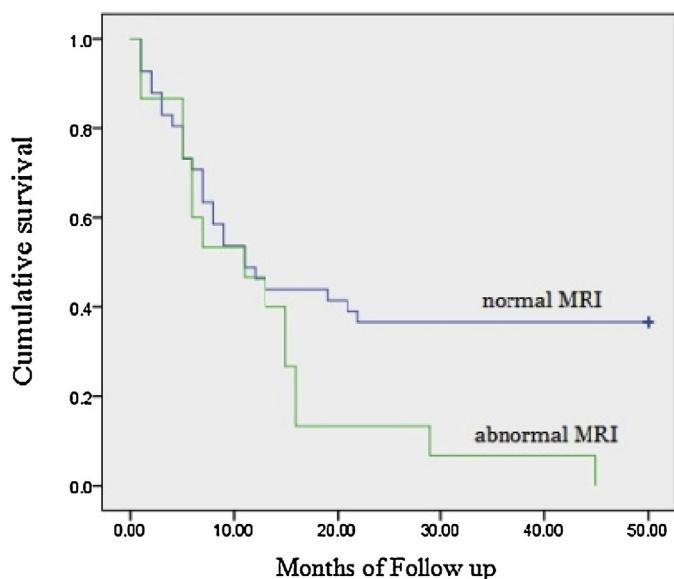


Fig. 6. Non-recurrence among patients with an abnormal VEEG that showed epileptiform discharges who had either a normal or an abnormal MRI scan.

Subgroup analysis found that the recurrence rates of generalized epileptiform discharges and partial epileptiform discharges were 68.8% and 75.0%, respectively, and were not significantly different ($p=0.735$). This may be due to a classification problem of the VEEG and might be the selection process of the patients. An abnormal MRI was not a predictive factor for seizure recurrence in the bivariate analysis of our study. However, an abnormal MRI was significantly associated with an increased risk of seizure recurrence in the VEEG group with epileptiform discharges abnormalities when comparing to those with normal MRI findings in the subgroup analysis. This finding merely showed that an abnormal MRI tended to relate to recurrence, which was consistent with other studies [10,11,23,24].

A first seizure with an abnormal EEG or brain imaging cannot be diagnosed as epilepsy unless there is sufficient evidence to substantiate a recurrence risk of more than 60% according to the practical clinical definition of epilepsy [5]. In our study, the recurrence risk associated with the first unprovoked seizure was more than 60% in patients with an epileptiform VEEG in a average follow-up of 2 years, similar to the 2-year risk for recurrence of 71% in children with epileptiform EEG in the Dutch Epilepsy Study [8], and them could be diagnosed as epilepsy.

Our study had some limitations. First, we recruited only 134 patients, and the ages ranged from 3 to 77 years, although our subgroup analyses found that age was not a predictive factor. Previous studies have always considered children and adults independently [10,12,14,15,17]. Second, the reoccurrence rate might be under estimate in our study that our follow-up time may be shorter than in other studies, with the longest follow-up time being 15 years [25]. Third, we can only according to the description of patients or witness to judge whether the event is more likely a epileptic seizure or not, and to classify the seizure type. Our researchers are well experienced in the field of epilepsy, however, some patients may be misdiagnosed. Fourth, are 24-hour VEEG really better than standard EEG in the predictive value associated with seizure recurrence in first unprovoked seizure patients? In general, recording the EEG for a longer time may provide more information for classifying the type of seizures or epilepsy. However, in our study, none of the patients had a seizure during the VEEG monitoring. The value of VEEG may solely be that in monitoring the EEG over a longer time period the rate of detecting abnormalities will improve.

5. Conclusions

An abnormal VEEG was a risk factor associated with seizure recurrence after a first unprovoked seizure. Especially, if the VEEG showed epileptiform discharges it was significantly associated with increased recurrence rates. The recurrence risk would meet the diagnostic criteria for the patients with a first unprovoked seizure and a VEEG with epileptiform discharge abnormality and could be used to diagnose epilepsy according to the practical clinical definition of epilepsy published by ILAE in 2014.

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Conflict of interest

None of the authors has any conflict of interest to disclose.

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